0580_w24_ms_41_Q5

Question	Answer	Marks	Partial Marks	Guidance
5(a)	$\frac{5}{8}$ oe	**1**		In all parts: Only accept fraction, percentage and decimal equivalents (3sf or better) Ignore incorrect cancelling or conversion following a correct answer. Penalty –1 once for 2sf or words e.g. 1 in 2 or 1 out of 2 **But not** ratio, not 1 to 2
5(b)(i)	Tree diagram correct probabilities on 3 pairs of branches $\frac{3}{8}$ $\frac{5}{8}$	**2**	**B1FT** for one pair of branches of first stage or second stage correct	FT *their* fraction in (a) dep on their pair of fractions adding to 1
5(b)(ii)	$\frac{17}{32}$ oe	**3**	**M2FT** for *their* $\left(\frac{3}{8}\times\frac{3}{8}\right)+\left(\frac{5}{8}\times\frac{5}{8}\right)$ oe or **M1FT** for one correct product seen	0.53125, 0.5312 to 0.5313, or 0.531 For method FT *their* tree diagram from (b)(i) but allow recovery to correct tree. For M2 or M1 must see product of their fractions. Not as part of a product of 3 or more fractions but accept eg $2\times\frac{3}{8}\times\frac{3}{8}$

0580_w24_ms_41_Q5

5(c)	$\frac{15}{56}$ oe	**3**	**M2 FT** for $\frac{3}{8}\times\frac{2}{7}\times\frac{5}{6}\times k$ where k is 1, 2 or 3 or **M1FT** for $\frac{3}{8}$ and $\frac{2}{7}$ and $\frac{5}{6}$ seen oe or for showing the 3 possible combinations If 0 scored, **SC1** for answer $\frac{135}{512}$ oe	0.268 or 0.2678 to 0.2679 Numerators could be in any order For M2 or M1 FT *their* 3/8 on first part of tree diagram from (b)(i) but allow recovery. Denominators must be decreasing by 1 and *their* 5 must be the numerator from *their* tree For M2 must show product of 3 fractions, eg $\frac{5}{56}$ with no working is M0 oe $\frac{5}{8}$ and $\frac{3}{7}$ and $\frac{2}{6}$, or $\frac{3}{8}$ and $\frac{5}{7}$ and $\frac{2}{6}$ e.g. BBR BRB RBB 0.264 or 0.2636 to 0.2637….